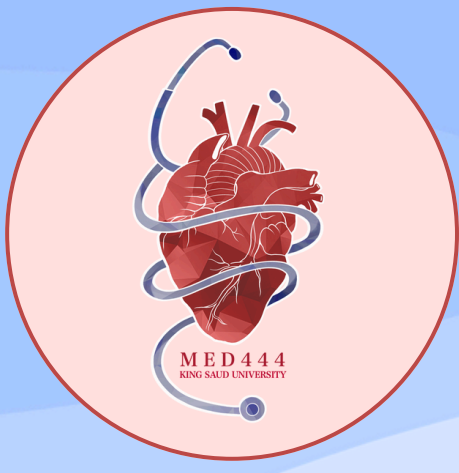




# Clinical data and big data analytics

## Editing files

- main text- male slides
- female slides- important<sub>1</sub>
- dr notes - textbook
- extra / old slides

# objectives



- Define data, information, and knowledge



- Types of clinical data



- Informatics vs. Information Technology and Computer Science



- Data to information



- Information to Knowledge



- Clinical Data Warehouse



- Use of Aggregated Clinical Data



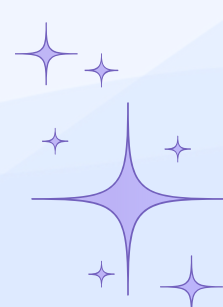
- What Makes Informatics Difficult?



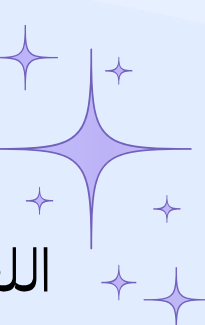
- Big data

The lecture presented through blackboard  
by:Dr.jwaher Almulhem

Click here to view the resources was attached with  
slides in blackboard



اللهم انفعني بما علّمتني، وعلمني ما ينفعني، وزدني علماً





# Data, Information, and Knowledge

1

**Data** are symbols or observations reflecting differences in the world.(e.g.5)

2

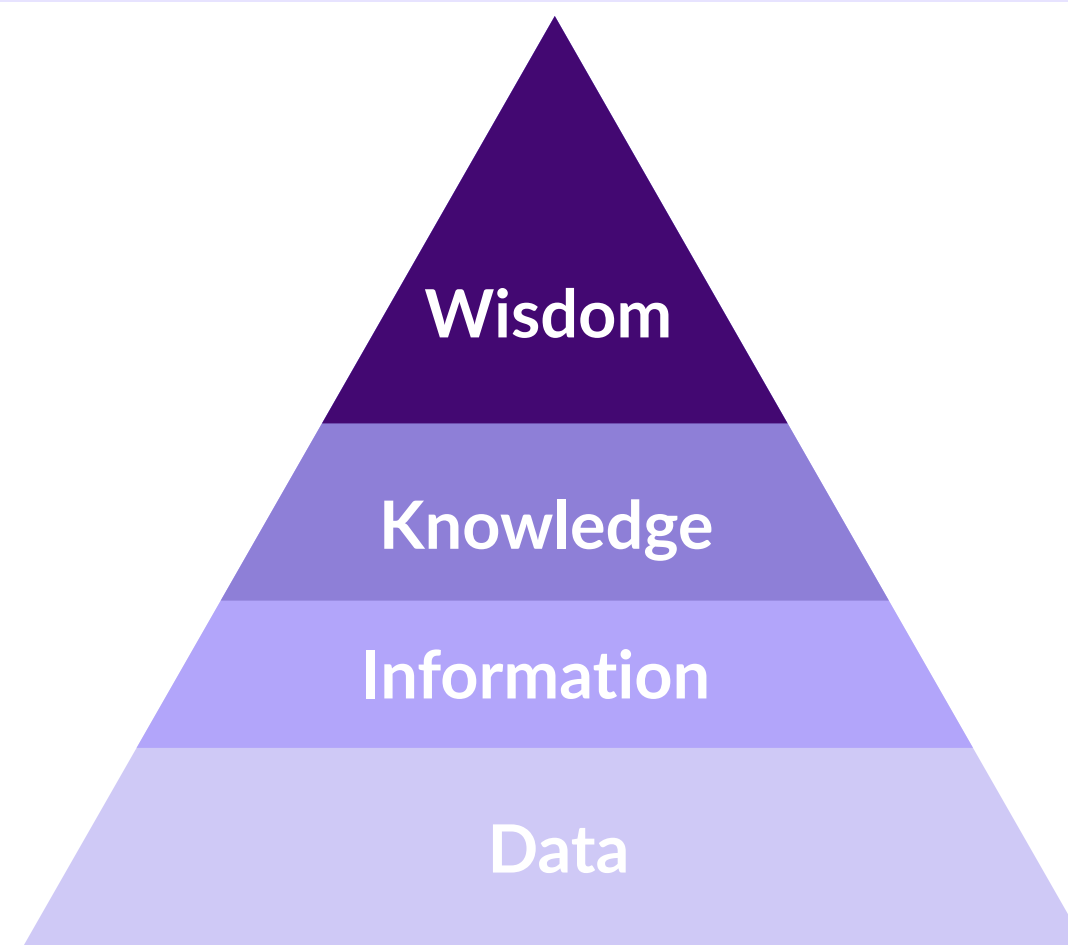
**Information** is data with meaning. (If we added the word “fingers” to number 5. “5 fingers”, now it has meaning)

3

**Knowledge** is information that is justifiably believed to be true. (elevation in FBG level, the likelihood of diabetes is increased.Smokers are likely to develop lung cancer more than non-smokers)

4

**Wisdom** is the critical use of knowledge to make intelligent decisions.



**Data**

Each zero or one is a bit  
Byte is a series of 8 bits

## Bits can occur as various data types

- Integers (numbers ex:14)
- Floating point numbers (decimals)
- Characters (A-Z)
- Character strings (ex: words)

Note that these data types do not define meaning. A computer does **not** “know” whether 3.14159 is a random number or the ratio of the circumference to the diameter of a circle (known as pi or  $\pi$ )



From textbook

## File format:

- image files (JPG, GIF, PNG),
- text files,
- sound files (WAV, MP3)
- video files (WMV, MP4)

Data are aggregated in many file formats, which makes sharing files possible

Common or standardized file formats allow sharing of files between computers and between applications. For example, as long as your digital camera stores photos as JPG files, you can use any program that can read JPG files to view your photos.



From textbook

it is important to recognize that neither data types nor file formats define the meaning of the data



# Clinical data

| Types of clinical data                    |                                                        |                                                                                           |
|-------------------------------------------|--------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Narrative                                 | Numerical measurements                                 | Coded Data                                                                                |
| recording by clinician- maternity history | blood pressure, temperature, glucose level, heart rate | selection from a controlled terminology system, <u>anything that has a drop down menu</u> |
| Textual Data                              | Recorded Signals                                       | Pictures                                                                                  |
| other results reported as text            | EKG, EEG                                               | radiographs, photographs, and other images                                                |

## General categories of data entry

Free-form entry by historical methods: writing, dictation, typing

Structured (menu-driven) data entry by mouse or pen

Speech recognition for either of above



**IMPORTANT!**

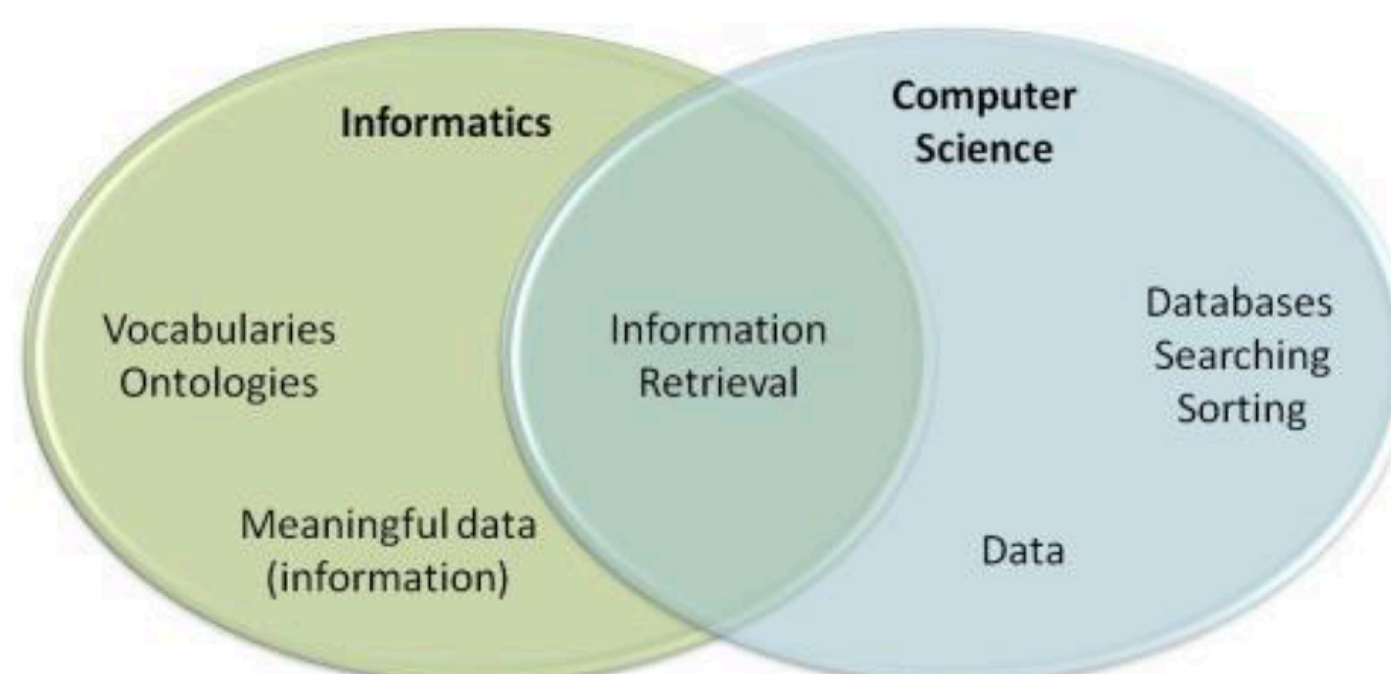
## Informatics and Computer Science

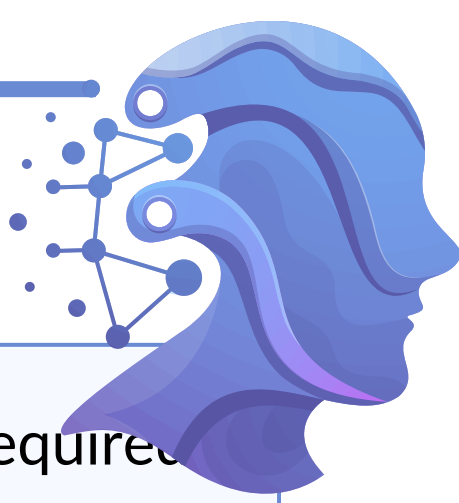
What is the domain of computer science? DATA (Meaningless)  
What is the domain of informatics? Information (Meaningful)

- **Data** are the domain of computer scientists and IT technologist, but information is the domain of **informatics** and informaticians (The meaning of the data is of secondary importance to computer scientists)

-so we can say that **computer science deals with meaningless data**

- **Information retrieval** involves both **computer science** (data) and **informatics** (information). Information retrieval is defined as finding material, (usually documents) of an unstructured nature, (usually text) that satisfies an information need by retrieving documents from large collections, (usually stored on computers).  
Example of information retrieval task: find the relationship between heart attacks and aspirin>all records will be retrieved.



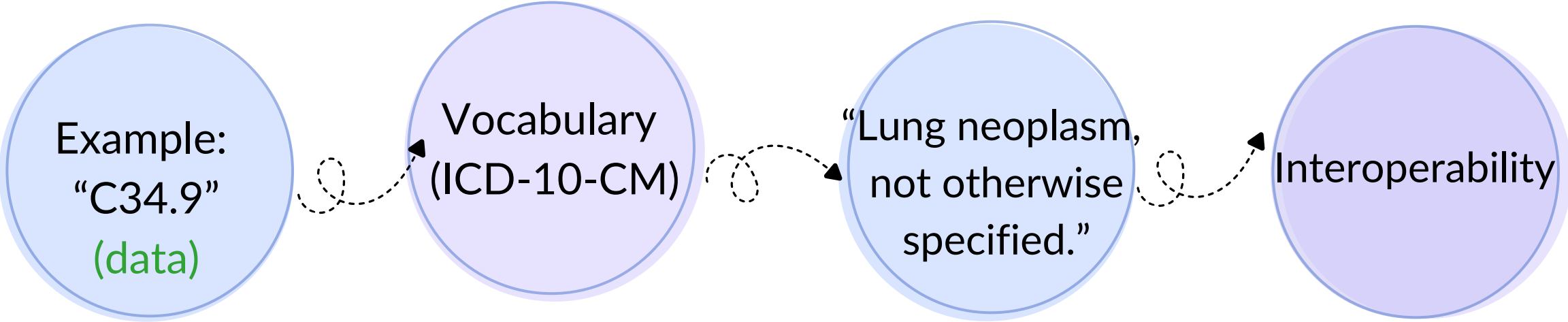


# Artificial Intelligence (AI)

- AI is concerned with the development of systems that can do something that previously required human intelligence.(chess ,driving vehicles)
- Example in medicine: Dermatologic (skin) lesion categorization.This system was developed by Stanford University, and it can categorize images of skin lesions to benign and malignant. This is done by training the system. Large data sets (images) are inserted with labels, (benign, malignant,...) and then data sets (images) are added without labels.
- The system tries to learn to be able to identify lesions.
- Limitations:
  - 1. It requires large sets of labeled data to train the system.
  - 2. Cannot justify judgment. (Why did it classify this picture as benign? )

## Data to Information

Interoperability:  
Transmission of information between computer systems.

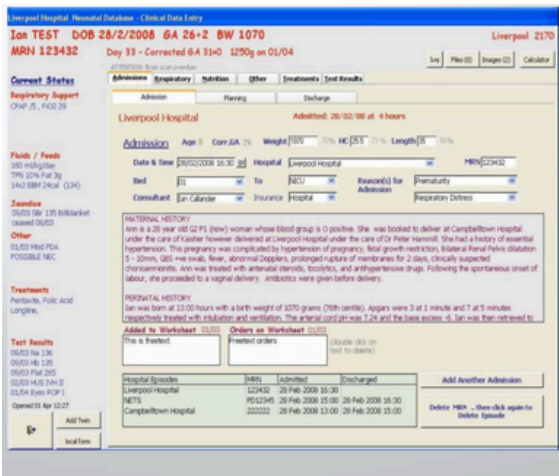


Explanation: C34.9 is code for Lung cancer in computers. However, this datum is of no meaning if we didn't link it to a known vocabulary. The computer still stores only data, no information.

ICD: International Classification of Disease, 10 is the version.  
Interoperability: ability of two or more systems or components to exchange the information and use the information that has been exchanged.(Ex: medical record at hospital A sends information to a medical record at hospital B without losing this information.)

## Information to Knowledge , 2 types of data

- 1 **Unstructured (free text)** such as discharge summaries or pathology reports.(Simple human language ex: history)
- 2 **Structured data** may include billing codes, lab results (e.g.sodium=140mg/dl)(from drop down menus such as billing codes, lab results, problem lists, and medication lists)



### Advantages and disadvantages

(ex: retrieval of all patients medical records that are in ICD-10-CM system with the same disease such as breast cancer. )

### Natural language processing (NLP)

It's used to make computers understand unstructured data. ( takes the natural human language as an input to process it. (Low-level processing: assigns the part of speech to words, “verbs, nouns,” while high-level processing such as Siri.)

|               | Structured data                             | Unstructured                                                     |
|---------------|---------------------------------------------|------------------------------------------------------------------|
| Advantages    | Much easier to manage                       | able to express anything                                         |
| Disadvantages | may not accurately reflect clinical reality | may be difficult to convey with a “one size fits all” vocabulary |

# Clinical Data Warehouse IMPORTANT!

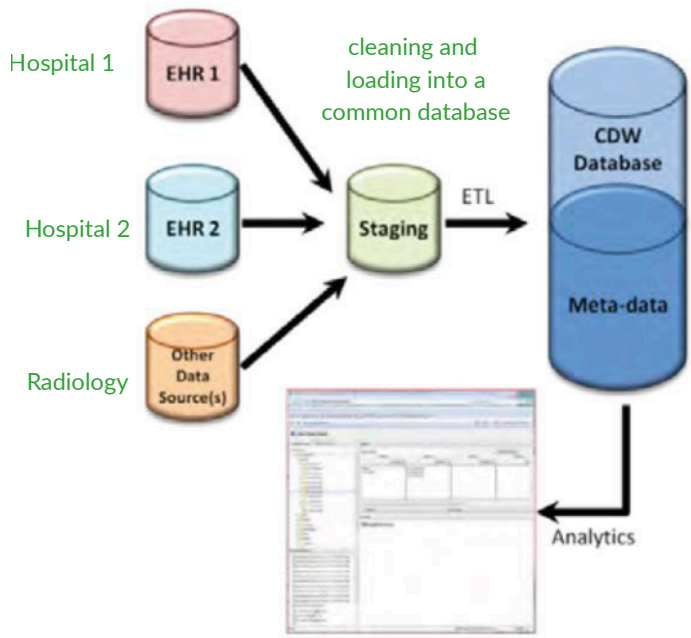


A modern way to convert medical information to knowledge is to use a clinical data warehouse (CDW).

|                             |                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition                  | a database system that collects, integrates and stores clinical data from a variety of sources including electronic health records (EHR), radiology and other information systems.                                                                                                                                                                                                                 |
| Staging                     | the process of cleaning the data from any incorrect or missing values.                                                                                                                                                                                                                                                                                                                             |
| Meta-Data                   | “data about data =data that describe other data example: ICD-10-CM<br>مثال لما حولنا حولنا use vocabulary<br>this is meta data احتاج أضيفها إلى clinical data warehouse عشان يعرف لي coded data<br>specific type of analytics organised بعملها بعد ما ادخل data وتكون<br>summary statistics ,count ,mean زي simple ممكن تكون<br>association between medication and disease زي Sophisticate او تكون |
| Difference between EHR& CDW | EHR: are systems implemented in hospitals that focus on <b>individual</b> data for one patient.<br>-Address real time updates regarding patient data<br>● CDW: support many EHR, focus on <b>group</b> query.<br>- Information update depends on how often it is updated (monthly, weekly,... unlike EHRs,they're designed to support real-time updating and retrieval of individual data          |

## Uses

If you want to retrieve all women who are 40 years or older what will you use ? CDWs



1

Monitor Quality by allowing users to query for specific quality measures. e.g. retrieve all women who are 40 years or older who have not had a mammogram in the past year

2

Identify trends.

3

Comparative effective research(CER): (is the treatment A better than treatment B? We can answer this question by looking for the prognosis of people who were treated with both in the system)

## i2b2 platform (an example of CDW) [\(link from slides\)](#)

Informatics for integrating biology and the bedside (i2b2): is a harvard study used by many other academic institutions in the US.

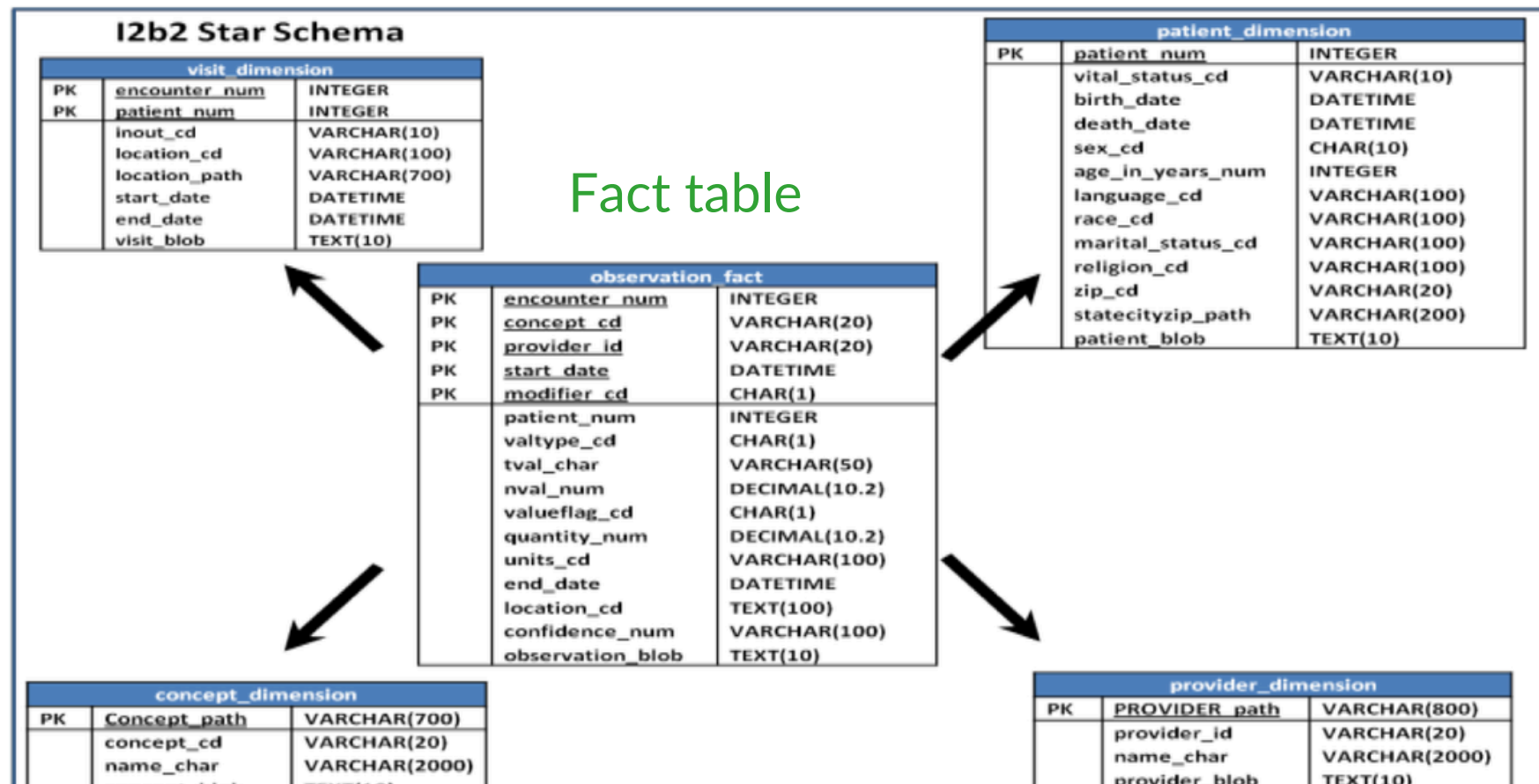
- Its an open source and modular and incorporate genomic and clinical information for research purposes.
- Database consists of **facts** (diagnoses, lab results, etc.) queried by users and dimensions that describe the fact.

**Design:** star scheme



## i2b2 platform

Dimensions  
table



In the middle there is a fact table (quantitative, information).

- On the side dimension tables (more information about these facts)
- Dimensions tables provide qualitative information that answer (where, when and what)

## Use of Aggregated Clinical Data

**Concept extraction:** the problem of identifying concepts within unstructured data, such as discharge summaries or pathology reports. غالباً يكون مربوط

بcontrol vocabulary

- Usually, these concepts are mapped to a controlled vocabulary.

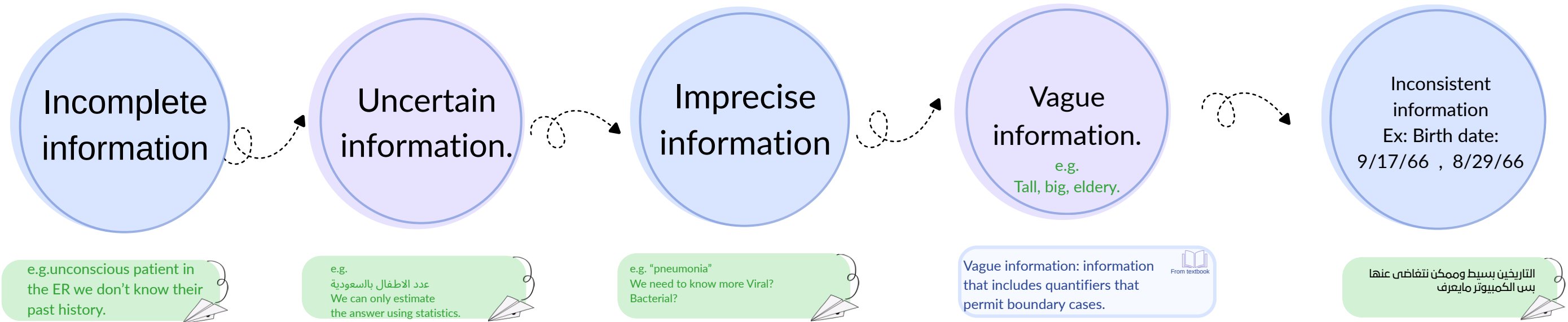
● **Classification:** the problem of categorizing data into two or more categories.

.-Supervised machine learning مثل سستم جامعة ستانفورد



# What Makes Informatics Difficult?

## Biomedical Informatics is Difficult:



## Big data

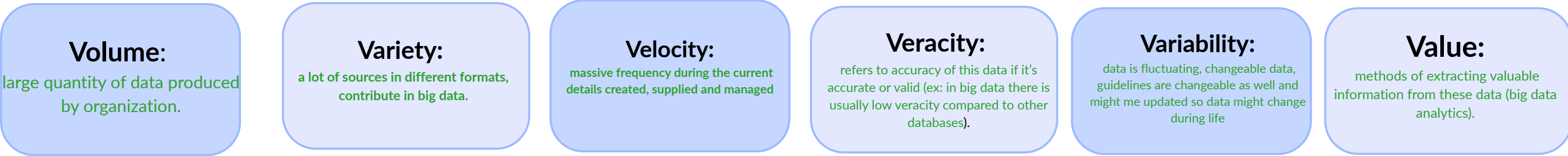
Definition: collecting large collections of data from various healthcare foundations followed by storing, managing, analyzing, visualizing, and delivering information for effective decision making.

As illustrated in the example, all these imperfections may be found in healthcare. And humans can deal with these imperfections but computers cannot.

### The Six V's of big data:

**IMPORTANT!**

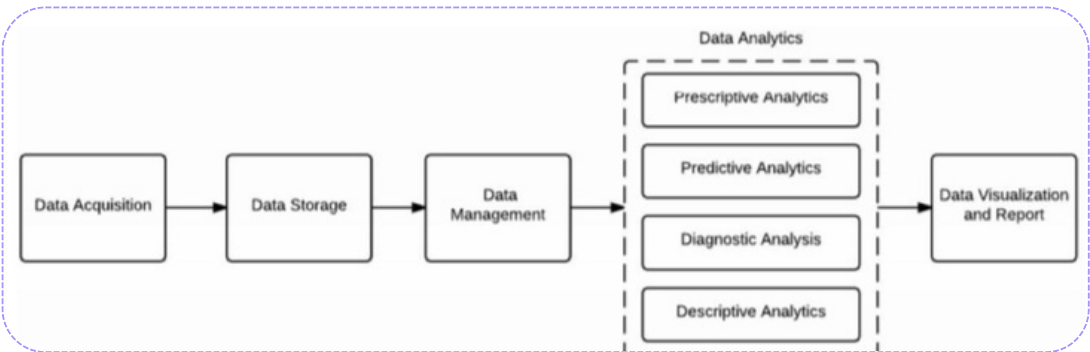
It's important to know the six V's and their definitions



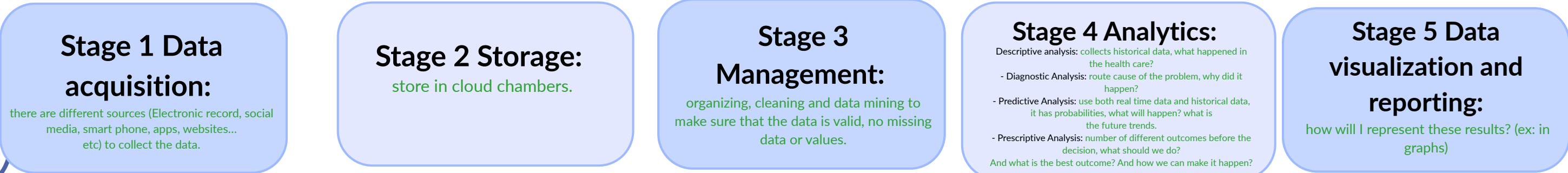
.Velocity includes both equivalent the rapidity of data manufacture and the rapidity of data handling to meet demand-  
-Big data has low veracity, it can never be 100% accurate, and it is difficult to validate. Since most of the data comes from unknown sources.

From textbook

- Organized (structured data: laboratory data, sensors data, data from databases).  
- Semi-organized: a combination of organized and unorganized (ex: website data and health records).  
- Unorganized "most common" (ex: social media data, medical records that doesn't use medical terminology).



**Analysis:**  
process of transforming raw data into information.





# Conclusion

- Users must be able to “make sense” of clinical data; to make clinical data meaningful (data → information) and then learn from aggregated clinical data (information → knowledge)
- Computer scientists focus on data, while informaticists focus on information
- The transformation of information into knowledge is a primary goal of informaticists
- Clinical data warehouses are increasingly used to research clinical questions and generate knowledge from information
- Sources of big data.

- Data are observations reflecting differences in the world (e.g., “C34.9”) while information is meaningful data or facts from which conclusions can be drawn and knowledge is information that is justifiably believed to be true
- Data are largely the domain of information technology (IT) professionals and computer scientists; information and knowledge are the domains of informatics and informaticians.
- Vocabularies help convert data into information
- The transformation of data into information and knowledge is a core concern of informaticians.
- When the real world, the conceptual model and the computational model match, we get useful answers from the computer
- Concepts relevant to health are relatively poorly defined compared to e.g. banking concepts
- There is a large “semantic gap” between health data and health information



From textbook

Extra

## Case Study: The Story of E-patient Dave

In January 2007, Dave deBronkart was diagnosed with a kidney cancer that had spread to both lungs, bone and muscles. His prognosis was grim. He was treated at Beth Israel Deaconess Medical Center in Boston with surgery and enrolled in a clinical trial of High Dosage Interleukin-2 (HDIL-2) therapy. That combination did the trick and by July 2007, it was clear that Dave had beaten the cancer. He is now a blogger and an advocate and activist for patient empowerment.

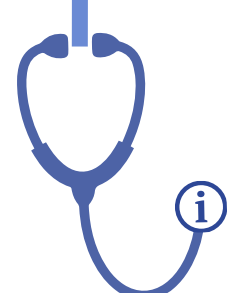
In March 2009, Dave decided to copy his medical record from the Beth Israel Deaconess EHR to Google Health, a personally-controlled health record or PHR. He was motivated by a desire to contribute to a collection of clinical data that could be used for research. Beth Israel Deaconess had worked with Google to create an interface (or conduit) between their medical record and Google Health. Thus, copying the data was automated. Dave clicked all the options to copy his complete record and pushed the big red button. The data flowed smoothly between computers and the copy process completed in only few moments.

What happened next vividly illustrated the difference between data and information. Multiple urgent warnings immediately appeared, including a warning concerning the prescription of one of his medications in the presence of low potassium levels (hypokalemia) (Figure 2.2). Dave was taking hydrochlorothiazide, a common blood pressure medication, but had not had a low potassium level since he had been hospitalized nearly two years earlier.

Worse, the new record contained a long list of deadly diseases (Figure 2.3). Everything that Dave had ever had was transmitted, but with no dates attached. When the dates were attached, they were wrong. Worse, Dave had never had some of the conditions listed in the new record. He was understandably distressed to learn that he had an aortic aneurysm, a potentially deadly expansion of the aorta, the largest artery in the human body.

Why did this happen? In part, it was because the system transmitted billing codes, rather than doctors’ diagnoses. Thus, if a doctor ordered a computed tomography (CT) scan, perhaps to track the size of a tumor, but did not provide a reason for the test, a clerk may have added a billing code to ensure proper billing (e.g., rule out aortic aneurysm). This billing code became permanently associated with the record. To put it another way, the data were transmitted from Beth Israel’s computer system to Google’s computer system quickly and accurately. However, the meaning of what was transmitted was mangled. In this case, the context (e.g., aortic aneurysm was a billing concept, not a diagnosis) was altered or lost. According to the definitions presented in chapter 1 (and reiterated later in this chapter), meaning is the defining characteristic of information as opposed to data.

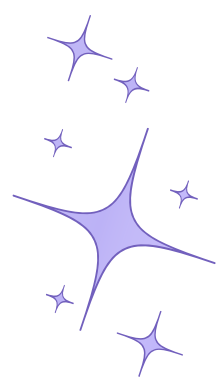
After Dave described what happened in his online blog<sup>2</sup> (<http://epatientdave.com/>), the story was picked up by a number of newspapers including the front page of the Boston Globe.<sup>3</sup> It also brought international attention to the problem of preserving the meaning of data. It became very clear that transmitting data from system to system is not enough to ensure a usable result. To be useful, systems must not mangle the meaning as they input, store, manipulate and transmit information. Unfortunately, as this story illustrates, even when standard codes are stored electronically, their meaning may not be clearly represented.



# Reference excerpts



- Information is meaningful data or facts from which conclusions can be drawn (e.g., ICD-10-CM code C34.9 = “Malignant neoplasm of unspecified part of bronchus or lung”).
- Knowledge is information that is justifiably believed to be true (e.g., “Smokers are more likely to develop lung cancer compared to non-smokers”).
- Why do we aggregate data into file formats?
  - 1 To specify the way the data are organized within the file.
  - 2 Common or standardized file formats allow sharing of files between applications.
- When converting data to information, one must consider the conceptual model to convert it to the computational model. Conceptual model:
- The difference between computer science and informatics :  
Computer science: computer scientists and IT professionals concentrate on technology, including computing systems composed of hardware and software as well as the algorithms implemented in such systems. For example, computer scientists develop algorithms to search or sort data (meaningless) more efficiently. Note that what is being sorted or searched is largely irrelevant. In other words, the meaning of the data is of secondary importance.
- Informatics: Information and knowledge (meaningful), are addressed by informatics. To an informatician computers are tools for manipulating information.
- What is information retrieval?  
Finding material (usually documents) of an unstructured nature (usually text) that satisfies an information need by retrieving documents from large collections (usually stored on computers).
- **Information retrieval is concerned with meaningful information NOT data, and it involves computer science and informatics.**
- AI or “machine learning” have enabled computers to solve problems that have previously resisted automation.
- What is deep learning?  
It's the use of multi-layer neural networks to learn patterns such as the features of objects in an image.
- Limitations of deep learning:
  - 1-It requires large sets of labelled data to “train” the system.
  - 2-The system cannot explain “why” it does something.
- Structured data vs Unstructured data  
Structured:
  - It may include billing codes, lab results (e.g., Sodium = 140 mg/dl), problem lists (e.g., Problem #1 = ICD-10-CM C34.9 = “Lung Neoplasm, Not Otherwise Specified”), medication lists.
- Structured data are much easier to manage and are computationally tractable.
- Unstructured:
  - Free text is simply human language such as English, called natural language.
  - Key portions of clinical notes are still often dictated and are represented in records as free text.
  - 90% of data are unstructured.
  - Has the advantage of being able to express anything that can be expressed using natural language.
  - Difficult for computers to process as it requires NLP.
- CDWs vs EHRs CDWs:
  - Variety of analytics can be applied on data, and the results presented to the user via a user interface. Examples of simple analytics include summary statistics such as counts, means, medians and standard deviations. More sophisticated analytics include associations (e.g., does A co-occur with B) and similarity determinations (e.g., is A similar to B).
  - Designed to support queries about groups. -CDWs are used to identify trends.
- EHRs:
  - Designed to support real-time updating and retrieval of individual data.
  - Databases that support EHRs are designed for efficient real-time updating and retrieval.
  - Semantic gap: A wide semantic gap makes informatics difficult because concepts relevant to health are relatively poorly defined. e.g. “sick” can be high BP or low BP or an aortic aneurysm or any kind of illness.



# SUMMARY



| Data                             |                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| definitions                      | <ul style="list-style-type: none"><li>• Data: are symbols or observations reflecting differences in the world.</li><li>• Information: is data with meaning.</li><li>• Knowledge: is information that is justifiably believed to be true.</li><li>• Wisdom: is the critical use of knowledge to make intelligent decisions.</li></ul> |
| Clinical data Types              | Narrative ○ Numerical measurements ○ Coded data ○ Textual data ○ Recorded signals ○ Pictures                                                                                                                                                                                                                                         |
| General categories of data entry | Free-form , Structured (menu-driven) data, Speech recognition                                                                                                                                                                                                                                                                        |
| Artificial intelligence (AI)     | AI is concerned with the development of systems that can do something that previously required human intelligence.                                                                                                                                                                                                                   |
| Data to information              | <ul style="list-style-type: none"><li>• ICD: International Classification of Disease, 10 is the version.</li><li>• Interoperability: Ability of two or more systems or components to exchange the information and use the information that has been exchanged.</li></ul>                                                             |

## Clinical Data Warehouse

|                                                                                                                            |                                                                                                                                                                                                         |                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Definition :</b><br>A modern way to convert medical information to knowledge is to use a clinical data warehouse (CDW). | <b>Function :</b><br>a database system that collects, integrates and stores clinical data from a variety of sources including electronic health records (EHR), radiology and other information systems. | <b>Uses :</b><br>Monitor Quality by allowing users to query for specific quality measures<br>Identify trends<br>Comparative effective research: practice based research, answers very specific questions |
|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

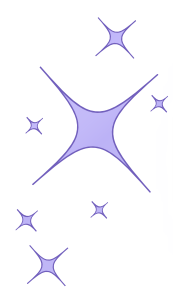
|                                 |                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use of Aggregated Clinical Data | <ul style="list-style-type: none"><li>• Concept extraction: the problem of identifying concepts within unstructured data, such as discharge summaries or pathology reports.</li><li>• Usually, these concepts are mapped to a controlled vocabulary.</li><li>• Classification: the problem of categorizing data into two or more categories.</li><li>• Supervised machine learning.</li></ul> |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## What Makes Biomedical Informatics Difficult?

|                                                                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Incomplete information.</li><li>• Uncertain information.</li><li>• Imprecise information.</li><li>• Vague information.</li><li>• Inconsistent information.</li></ul> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Big data

|                                                                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Definition:</b> collecting large collections of data from various healthcare foundations followed by storing, managing, analyzing, visualizing, and delivering information for effective decision making. |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



# Doctor's questions



**Q1:What role does clinical data play in research?**

reduces errors and makes data trustworthy, so that the analysis can be done more quickly.

**Q2:What is a clinical data warehouse, and how does it differ from an Electronic Health Record (EHR) system?**

clinical data warehouse is a database system that collects, integrates and stores clinical data from a variety of sources including electronic health records, radiology and other information systems.

EHRs are hospital-implemented systems that focus on a single patient's personal information.

So CDW differs in that it supports several EHRs and concentrates on group query.

**Q3:How can big data analytics aid in early disease detection and prevention?**

1- it makes it possible to identify patterns that increase the risk for a given disease and also determine new biomarkers for detection

2- big data analytics can predict an individual's risk of developing certain diseases

3- helps in Monitoring and surveillance (tracking outbreaks)

**Q4:What are the challenges associated with using big data in healthcare?**

Updating Data, Storage, Privacy, Security

**Q5:What are some common methods used for aggregating clinical data?**

EHR,Wearables,Claims Data

**Q6:What are some the primary sources of data that are aggregated in clinical settings?**

- 1.Medical chart abstraction
- 2.Electronic health records
- 3.Institutional or organizational databases
- 4.Death and birth records
- 5.Radiology information system

**Q7: What is the future potential of big data in healthcare?**

- application of machine learning and artificial intelligence over big data providing insights for improving the delivery of healthcare.

- This vast information base has the potential to drive innovative advances in patient care

- Facilitate research and develop pharmacology

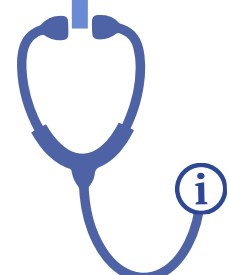
**Q8: what are the uses of AI in healthcare? mention two examples**

-risk prediction-Drug discovery-Clinical trials

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لا حول ولا قوة الا بالله

سبحان الله





**Q1: Finding material of unstructured nature that satisfies an information need by retrieving documents from large collections?**

A- Artificial intelligence

B- Information retrieval

C-Health informatics

**Q2: big data has low?**

A-velocity

B-Veracity

C-Variety

**Q3: Which of the followings is picture data?**

A-ECG

B-Blood test

C-X-ray

**Q4:Data that describe other data?**

A-Meta-Data

B-Big data

C-Structured data

**Q5:Which of the following is NOT one of the six V's of Big Data?**

A. Volume

B. Velocity

C. Validity

**Q6: Which stage of data analytics answers the question: “Why did it happen?”**

A. Descriptive analysis

B. Diagnostic analysis

C. Predictive analysis

استغفرالله واتوب اليه

لا حول ولا قوة الا بالله

سبحان الله

**Answer key:**

**Q1: B , Q2: B , Q3: C , Q4: A ,Q5: C , Q6: B**



# Team Leaders

Shahad Alshehri



Khalid Alghamdi

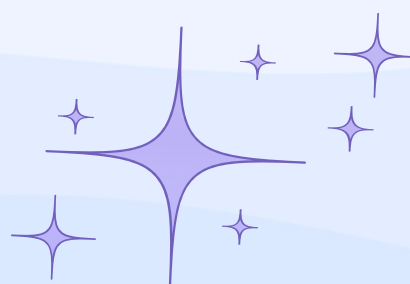


# Team Members

Lina Albaqiyah



Wasaif Alotaibi



سبحان الله، الحمد لله، لا إله إلا الله، الله أكبر،  
لا حول ولا قوة إلا بالله

